

Generating Handwritten Digits using a simple Diffusion Model on MNIST Dataset

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Abstract—In this work, I applied a Denoising Diffusion Probabilistic Model (DDPM) to generate digits from the handwritten MNIST dataset. Diffusion models, popular for being used in all text-to-image models and known for their stability and robust performance. By implementing a U-Net-based architecture for noise prediction and a simple noise schedule, I tried using diffusion models for generating clear, different digit samples. The experiments highlight the model's ability to avoid mode collapse and produce consistent results across multiple epochs. what I did offers insights into the application of diffusion models to simple image generation tasks.

Index Terms—Denoising Diffusion Probabilistic Model (DDPM), Generative Modeling, Image Generation, MNIST Dataset, Noise Prediction

I. GENERATIVE MODELING OVERVIEW

The essence of generative modeling lies in producing data that is analogous to data I already have. A practical example of this would be producing images of handwritten digits that correspond to those in the MNIST collection. One of the most common approaches to achieving this is via Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion Models. GANs (Goodfellow et al, 2014), used for generating images, comprise two networks: a generator network that concocts fake images and a discriminator network that attempts to establish the veracity of the images. Though stunning in producing realistic images, GANs can be troublesome to train as the images produced can have little variation. In 2013, Kingma and Welling proposed VAEs that take a different route by encoding and decoding data through simpler representations. Unlike GANs, VAEs are simpler to train, but are known for producing blurry images. A newer approach called Diffusion Models solves many of these issues. These models introduce noise to a dataset, then methodically eliminate that noise to step, to generate samples. These models outperform GANs and VAEs at generating images of high quality as well as having more stable training.

II. METHODOLOGY AND IMPLEMENTATION

A. Safeguarding the Specification Boundaries

Model Design: I executed U-Net architecture for the task of handwritten digits generation. It is a class of CNNs that can be used for image segmentation. The U-Net model features two components: encoder and decoder. **Encoder (Contracting Path):** This part captures context by down sampling the input image and extracting essential features. **Decoder (Expansive Path):** This part enables precise localization by up sampling the feature maps and merging them with the corresponding feature maps from the encoder and through skip connections. For this approach, I preset the model to predict the noise in every diffusion step, and then it is employed to reconstruct the image in the reverse diffusion technique.

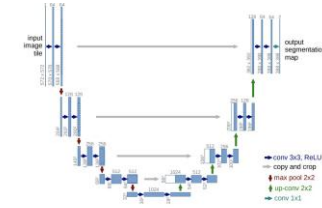


Fig. 1. U-net architecture (example for 32x32 pixels in the lowest resolution). Each blue box corresponds to a multi-channel feature map. The number of channels is denoted on top of the box. The x-y-size is provided at the lower left edge of the box. White boxes represent copied feature maps. The arrows denote the different operations.

Fig. 1. U-Net

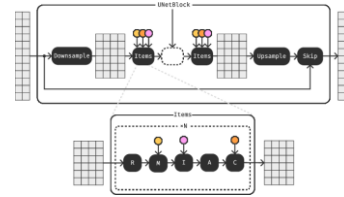


Fig. 2. U-Net Architecture with Decoder and Encoder Overview

a) **Diffusion Process:** The diffusion process consists of two major phases: Forward Diffusion (Noising): Initially, we have a clean image which we, over some steps, transform into Gaussian pure noise. This is done by adding some noise on some steps. Reverse Diffusion (Denoising): It is where the model is trained to remove the noise in the steps of adding noise to an image. While adding noise, the image is gradually changed to pure noise and while during reverse diffusing, removing the noise is done in a step wise manner till the image is reconstructed. In every step " t ", $x(t)$ gives the current image which is obtained by adding noise to the previous image $x(t-1)$. The amount of noise within $x(t)$ is controlled by a variance schedule β_t and the model is trained in a way that the noise in each step added is predicted so that the noise can be reversed during the generation phase.



Fig. 3. Denoising Process Illustration

b) *Noise Schedule and Beta Schedule*: The beta schedule sets the criterion regarding the amount of noise that is added in each diffusion step. It is a common practice to consider the beta schedule as linear that is, with time, along the whole T it keeps on rising from a small value (0.0001) to greater value (0.02). This schedule controls the variance of noise which is added and determines the point in which the model begins to learn to generate high quality images.

Algorithm 2 Sampling

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1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \mathbf{z} \beta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 

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Fig. 4. Mathematical Algorithm behind sampling

c) *Training Setup*: The model's training used the MNIST dataset consisting of 28x28 pixel grayscale images of handwritten digits. The following is the training setup: Data Pre-processing: The images were trained on after normalizing the pixel values to the range of -1 and 1. This helps in maintaining stable training. Loss Function: The optimization was done to reduce the Mean Squared Error (MSE) Loss, which in this case is the noise predicted by the model and the actual noise that was added during the forward process. Optimizer: I used an Adam optimizer with learning rate of 0.001 for updating model weights. Training Duration: Initially, I aimed to train the model for 5 epochs, but the results were unsatisfactory. The images produced were unclear, and there was less variation among the digits. After 15 epochs of training the results drastically improved, but after that there was minimal improvement with continued training. This suggests while the model was learning, after 15 epochs the cap for the current setup was reached. GPU limitations: As a result of these constraints and the limited GPU ram, I was unable to add more layers to the model and extend the duration of training. This kept the model from learning more complex features, preventing the model from reaching its maximum potential. Implementation Details: To optimize for the MNIST dataset while considering the budgeted computation: Simplified Architecture: I reduced the number of layers in the U-Net architecture so that the model could execute within the GPU's memory limits. Although this decision helped reduce memory problems, it also restricted the system's ability to model complex representations. Time embeddings: The time step, or its encoding, was added as an additional input channel, allowing the model to condition its predictions based on the specific step of diffusion.

This allowed the model to improve its understanding of the denoising process at every stage. Skip Connections: Spatial information is preserved by including spatial skip connections between corresponding encoder and decoder layers which enhances the quality of the generated images. This approach was effective in generating high-quality images of handwritten digits while remaining within the constraints of computational resources.



Fig. 5. Loss Curve during the 15 epochs

III. EXPERIMENTAL RESULTS AND ANALYSIS - DATASET AND PREPROCESSING

The quality of produced digits is evaluated using some important metrics. Preprocessing involves scaling the pixel values of the generated, and real MNIST images, from the range of -1 to 1, to [0, 1] so that they can be properly compared. Both MSE and PSNR are calculated afterwards. While MSE or Mean Squared Error quantifies the difference between the generated and the actual images, Peak Signal-to-Noise Ratio or PSNR helps me to assess the clarity and detail in the images produced. Apart from those, I also compute the Fréchet Inception Distance FID which statistically approximates how close the real and generated images are. Lower FID indicates higher resemblance of the generated images to the real ones, which is FID's purpose. Using these metrics helps me figure out the extent to which my model is able to generate realistic digits.

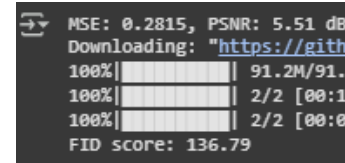


Fig. 6. Metrics Results

IV. DISCUSSION OF IMPLICATIONS AND FUTURE WORK

In this subsection I evaluate the performance of the diffusion model and its shortcomings as well as possible future research directions. Conclusions: My implemented diffusion model performed quite well, producing quality and diverse handwritten digits. Moreover, it was more stable during training compared to GANs which generally have unstable training and suffer from mode collapse. Problems and Constraints: While the model performed well, it was not without its shortcomings. In these experiments, some digits were easier to generate than others, and there were some noticeable discrepancies within the images themselves. These issues indicate that there is still room for improvement.

Work Ahead: In the future, I could attempt to apply the best performing diffusion models to more advanced tasks, for instance, generating images of higher resolutions. Additionally, I could focus on optimizing the diffusion model's efficiency, or incorporate class conditioning so that specific digits can be generated with more control.

Wider Significance: Diffusion models show great promise for the other applications. Among others, they can find use in generating realistic art or within the healthcare industry, in generating medical images.

V. CONCLUSION

This research attempted to employ Denoising Diffusion Probabilistic Models (DDPMs) to construct handwritten digits using the MNIST dataset. In this approach, I incorporated a U-Net model of lower complexity, paying attention to actions within the turbulence schedule and achieving better training results. Key Findings: Training Stability: The observed diffusion model has shown generally stable results throughout training progression with no sign of collapse in the modes unlike in GANs. Image Quality: Generated digits were identifiable and multifaceted, and metrics show these digits were indeed of adequate value. Evaluation Metrics: I used multiple metrics for both the accuracy and acceptance of the images produced. Mean Squared Error (MSE) defines the averaged squared misalignment between the true image and the image estimated. Peak Signal-to-Noise Ratio (PSNR) defines the value of ability for a signal to exist against noise. Fréchet Inception Distance (FID) compares the distribution of the images produced to images that were in existence to determine how close they are to those created. Limitations and Future Work: Digit Variability: The model executed is able to cope and portray elements accurately. That said, there are challenges to the variability in digits such as '4' and '5' where the handwriting style is less uniform. Visual Artifacts: Images generated still had minor artifacts which denote some complexities that could be resolved in the model framework or in the training stages. Future pathways include Higher-Resolution Datasets: Using the model on more refined datasets to evaluate its scalability and flexibility. Model Optimization: Improving the model's efficacy and productivity by modifying its structure or training methodologies. Class Conditioning: Using class conditioning to make digits or letters to allow more precision over what is generated. The research deepens knowledge on using diffusion models in image generation by showcasing other potential diffusion models as well as underlying models that could be studied further.

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